Supplementary Material

BD-LW: a leakage-aware image dataset and benchmark for traditional Bangladeshi weapon recognition

This document contains the detailed statistics that support the main text. All values come from the same runs, folds and locked test set as the main paper; E1, E2 and E3 denote training on all, single and multi images.

S1 Annotation guideline

Every image carries two labels: its weapon class and its subset (single or multi). Both were assigned according to a written annotation guideline that defines each class by the shape and use of the object rather than by its appearance in a particular photograph. *Churi* covers one-handed blades with a handle, from slim kitchen knives to broad cleavers; *Dao* covers long, heavy blades that curve and widen towards the tip; *Khor* covers razors whose blade folds into the handle; *Kodal* covers a broad, flat blade fixed at a right angle to a long handle; *Kural* covers a wedge-shaped head mounted on a handle; and *Shabol* covers a long, solid iron bar with a pointed or flattened end. The guideline also fixes how borderline cases are handled. An object was counted when its defining part (the blade, head or working end) was visible, even if the rest was hidden by the hand or by other objects. An image was labelled *single* when it showed exactly one such object and *multi* when it showed two or more objects of the same class. Images containing weapons of different classes, and images in which no defining part was visible, were not included.

A second annotator relabelled a stratified random sample of 300 images (25 per class and subset) with the same guideline; the agreement and all other validation checks are given in Table S17.

S2 Supplementary figures

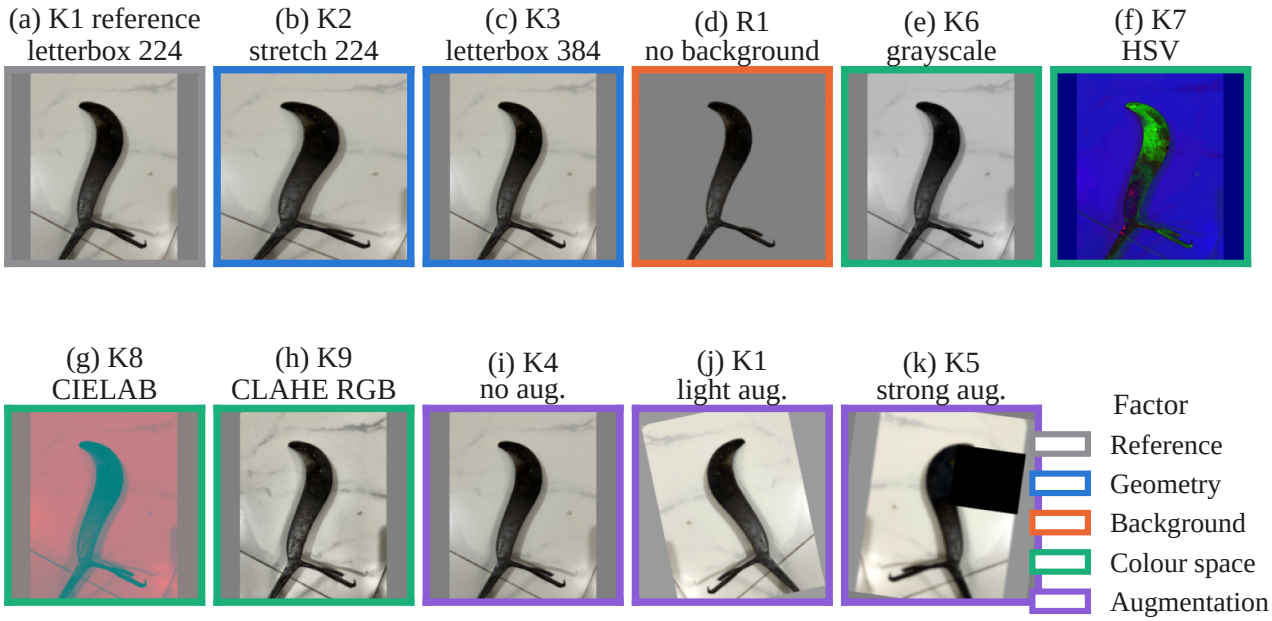


Fig. S1 Preprocessing variants applied to a representative BD-LW image: (a) the reference K1, (b)–(i) and (k) one-factor changes, and (j) the light training augmentation of the reference. Border colours denote the changed factor: geometry (blue), background (orange), colour space (green) and augmentation (purple); the reference is framed in grey.

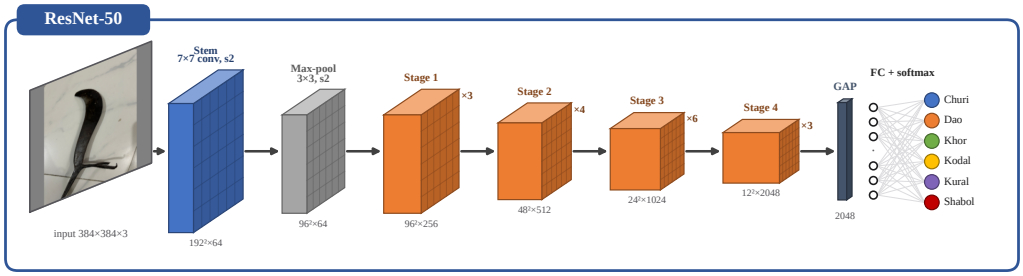


Fig. S2 Architecture of ResNet-50: 7×7 convolutional stem, four stages of bottleneck blocks and a six-class output head.

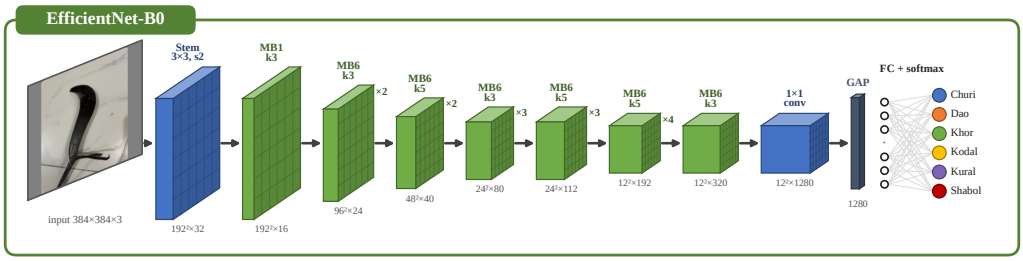


Fig. S3 Architecture of EfficientNet-B0: seven stages of mobile inverted bottleneck (MB) blocks with squeeze-and-excitation and a six-class output head.

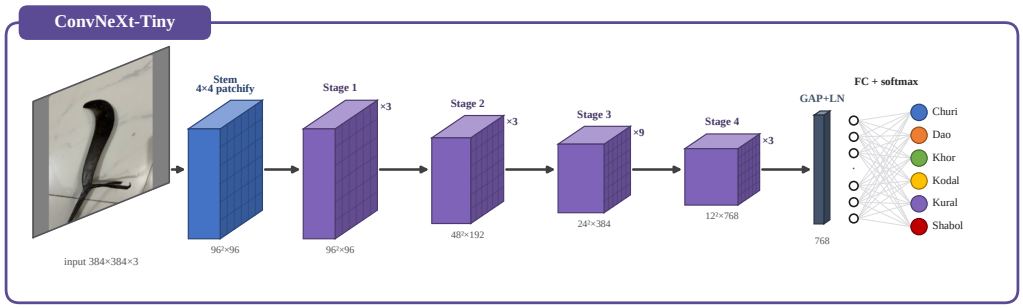


Fig. S4 Architecture of ConvNeXt-Tiny: 4×4 patchify stem, four stages of ConvNeXt blocks with 7×7 depthwise convolutions and a six-class output head.

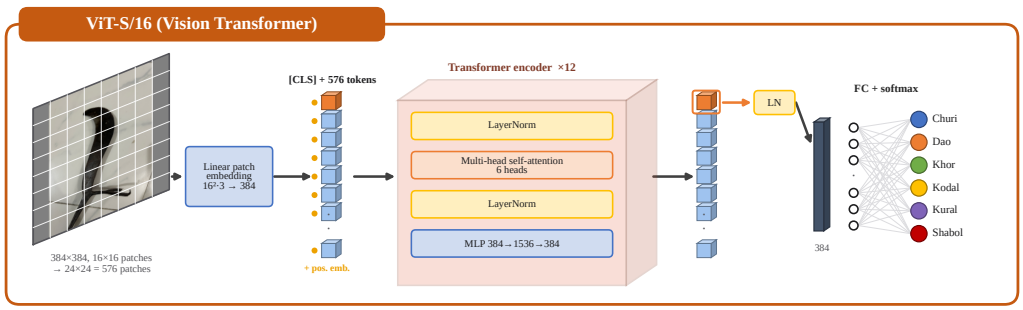


Fig. S5 Architecture of ViT-S/16: 16×16 patch embedding with class token and position embeddings, twelve transformer encoder blocks and a six-class output head.

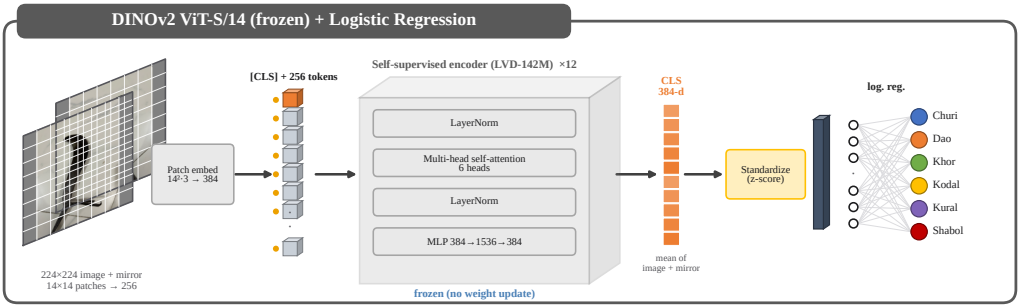


Fig. S6 Architecture of the DINOv2 baseline: frozen ViT-S/14 encoder with flip-averaged class-token features and a logistic regression classifier.

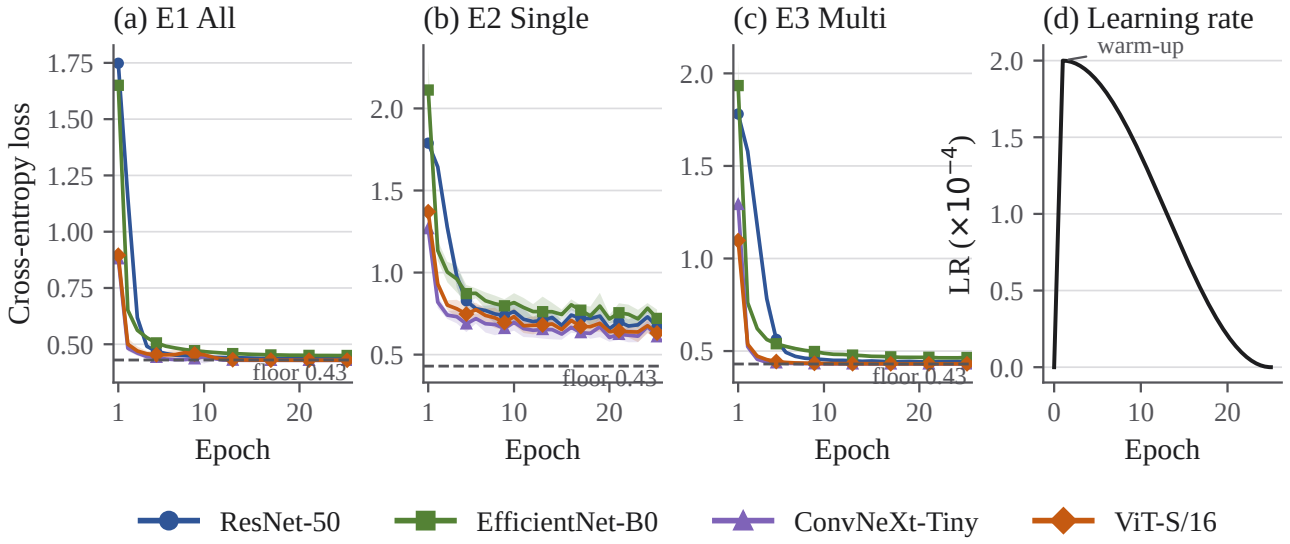


Fig. S7 Cross-entropy training loss over 25 epochs with recipe F for (a) E1 All, (b) E2 Single and (c) E3 Multi (mean ± SD over five folds; dashed line: label-smoothing floor of 0.43), and (d) the cosine learning-rate schedule with one-epoch warm-up.

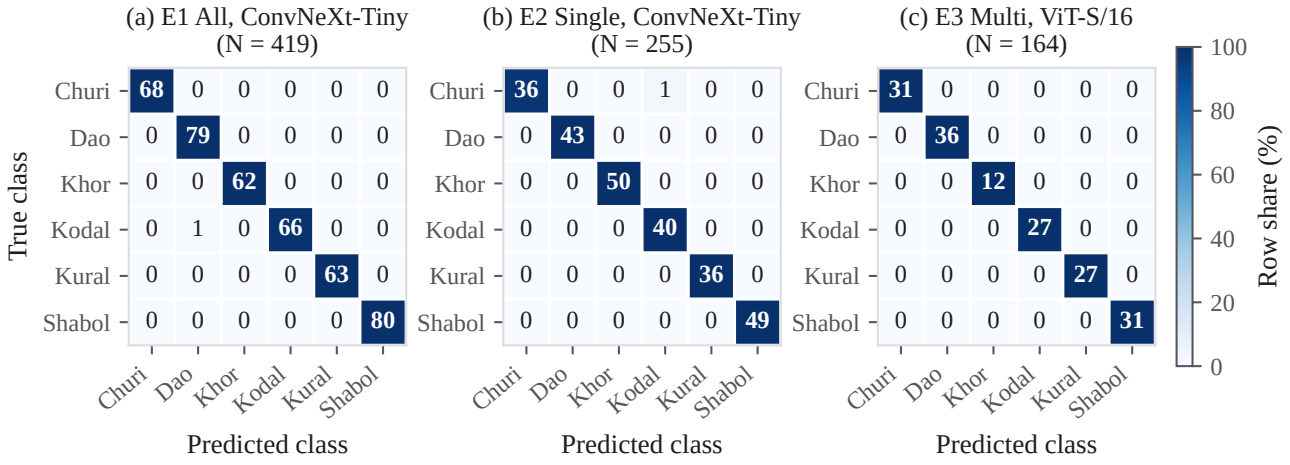


Fig. S8 Test confusion matrices of the best model in each scenario: (a) ConvNeXt-Tiny in E1 ($N = 419$), (b) ConvNeXt-Tiny in E2 ($N = 255$) and (c) ViT-S/16 in E3 ($N = 164$), all with recipe F. Cells give image counts, including zeros; shading gives the share of each true class.

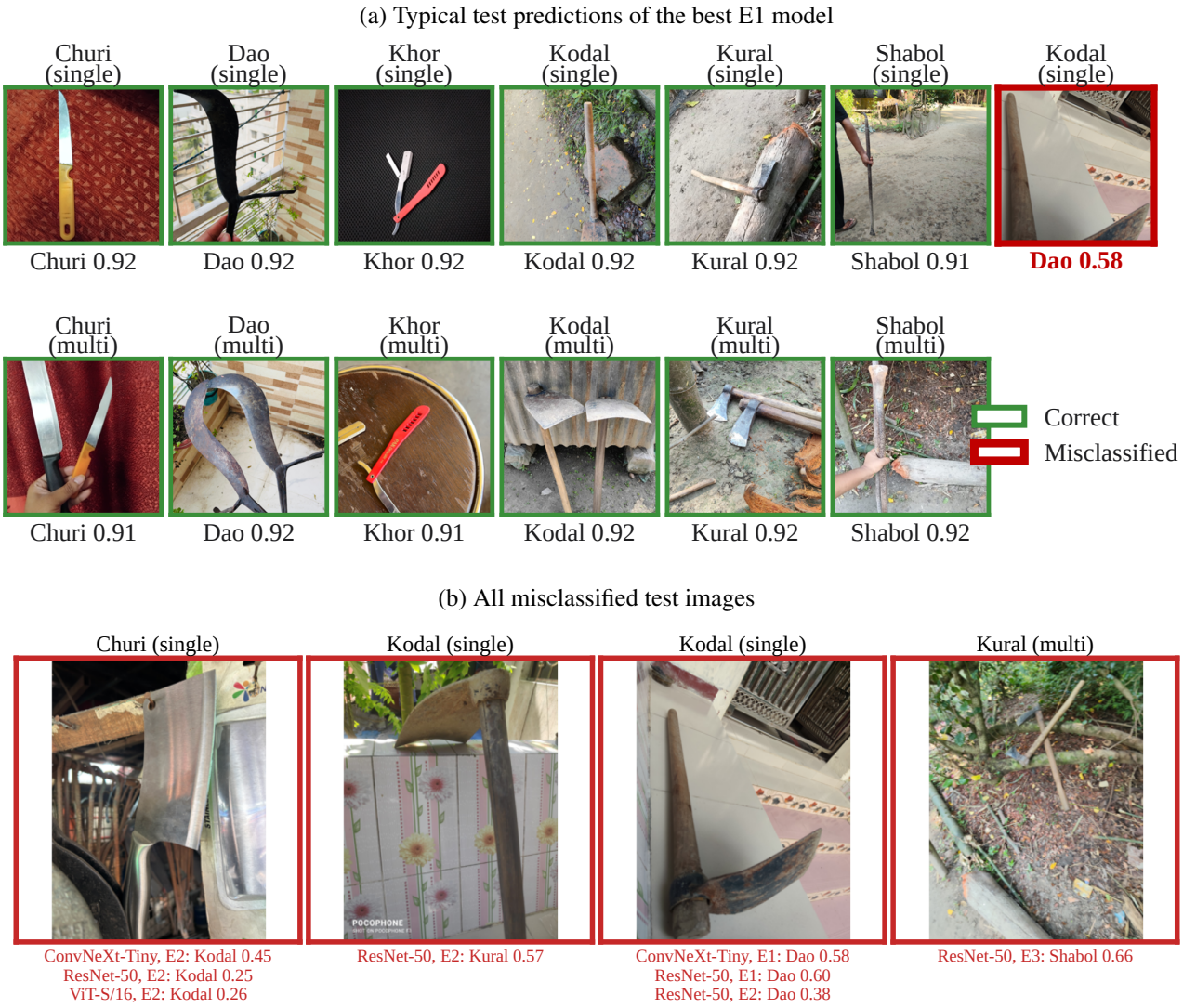


Fig. S10 Test predictions with recipe F. (a) ConvNeXt-Tiny in E1, with the true class above and the predicted class and confidence below each image; green and red borders mark correct and misclassified images. (b) Every test image misclassified by any of the four fine-tuned models in E1, E2 or E3 (eight errors on four images), with the true class and subset above and, below, each model and scenario that misclassified it, with the predicted class and its confidence.

S3 Supplementary tables

Table S1: ResNet-50 preprocessing ablation in each scenario: CV macro-F1 (mean ± SD over five folds), test macro-F1, difference from the reference K1 or R1 with 95% confidence interval, number of folds in which the setting was better, and Holm-adjusted p (paired t-test).

Pipeline	Change	CV macro-F1	Test	Δ	95% CI	Better	p_{Holm}
E1 All							
K1	reference	0.995 ± 0.003	0.995	–	–	–	–
K2	size: stretch 224	0.996 ± 0.003	0.998	+0.002	[-0.001, +0.004]	4/5	0.864
K3	size: letterbox 384	0.998 ± 0.002	0.998	+0.003	[-0.001, +0.007]	4/5	0.864
K4	augmentation: none	0.986 ± 0.004	0.993	-0.009	[-0.012, -0.006]	0/5	0.015
K5	augmentation: strong	0.994 ± 0.005	0.998	-0.001	[-0.008, +0.006]	1/5	1.000
K6	color: grayscale	0.986 ± 0.004	0.995	-0.009	[-0.015, -0.003]	0/5	0.154
K7	color: HSV	0.987 ± 0.002	0.992	-0.008	[-0.012, -0.004]	0/5	0.073
K8	color: LAB	0.991 ± 0.004	0.993	-0.004	[-0.008, +0.000]	1/5	0.654
K9	color: CLAHE	0.993 ± 0.003	0.993	-0.002	[-0.006, +0.002]	1/5	0.864
R1	reference	0.954 ± 0.004	0.974	–	–	–	–
R2	size: stretch 224	0.960 ± 0.004	0.969	+0.005	[-0.003, +0.013]	4/5	0.864
R3	size: letterbox 384	0.962 ± 0.007	0.981	+0.008	[+0.003, +0.012]	5/5	0.133
R4	augmentation: none	0.946 ± 0.011	0.974	-0.009	[-0.023, +0.005]	1/5	0.864
R5	augmentation: strong	0.952 ± 0.010	0.967	-0.003	[-0.017, +0.011]	3/5	1.000
R6	color: grayscale	0.929 ± 0.007	0.955	-0.026	[-0.034, -0.018]	0/5	0.015
R7	color: HSV	0.920 ± 0.014	0.950	-0.035	[-0.053, -0.016]	0/5	0.091
R8	color: LAB	0.940 ± 0.013	0.972	-0.014	[-0.032, +0.004]	1/5	0.864
R9	color: CLAHE	0.955 ± 0.005	0.962	+0.001	[-0.005, +0.006]	3/5	1.000
E2 Single							
K1	reference	0.996 ± 0.007	0.992	–	–	–	–
K2	size: stretch 224	0.995 ± 0.007	0.992	-0.001	[-0.003, +0.001]	0/5	1.000
K3	size: letterbox 384	0.996 ± 0.009	0.983	-0.000	[-0.003, +0.003]	1/5	1.000
K4	augmentation: none	0.990 ± 0.012	0.987	-0.006	[-0.013, +0.000]	0/5	0.450
K5	augmentation: strong	0.996 ± 0.005	0.987	+0.000	[-0.003, +0.003]	1/5	1.000
K6	color: grayscale	0.984 ± 0.008	0.979	-0.012	[-0.016, -0.008]	0/5	0.020
K7	color: HSV	0.988 ± 0.007	0.996	-0.008	[-0.011, -0.005]	0/5	0.019
K8	color: LAB	0.993 ± 0.006	0.983	-0.003	[-0.006, -0.001]	0/5	0.302
K9	color: CLAHE	0.992 ± 0.006	0.983	-0.004	[-0.006, -0.001]	0/5	0.302
R1	reference	0.951 ± 0.008	0.964	–	–	–	–
R2	size: stretch 224	0.956 ± 0.005	0.963	+0.005	[+0.001, +0.010]	5/5	0.312
R3	size: letterbox 384	0.964 ± 0.008	0.971	+0.013	[-0.003, +0.029]	4/5	0.544
R4	augmentation: none	0.943 ± 0.012	0.963	-0.008	[-0.023, +0.007]	1/5	0.818
R5	augmentation: strong	0.958 ± 0.009	0.951	+0.007	[-0.001, +0.016]	4/5	0.544
R6	color: grayscale	0.924 ± 0.005	0.941	-0.027	[-0.040, -0.013]	0/5	0.075
R7	color: HSV	0.912 ± 0.012	0.912	-0.038	[-0.050, -0.026]	0/5	0.015
R8	color: LAB	0.937 ± 0.014	0.954	-0.013	[-0.021, -0.005]	0/5	0.116
R9	color: CLAHE	0.943 ± 0.008	0.952	-0.007	[-0.016, +0.001]	1/5	0.544
E3 Multi							
K1	reference	0.996 ± 0.002	0.989	–	–	–	–
K2	size: stretch 224	0.995 ± 0.005	0.995	-0.001	[-0.006, +0.004]	1/5	1.000
K3	size: letterbox 384	0.993 ± 0.004	1.000	-0.003	[-0.010, +0.004]	1/5	1.000
K4	augmentation: none	0.991 ± 0.009	0.989	-0.005	[-0.015, +0.004]	1/5	1.000
K5	augmentation: strong	0.994 ± 0.003	1.000	-0.002	[-0.005, +0.002]	1/5	1.000
K6	color: grayscale	0.992 ± 0.006	0.988	-0.005	[-0.011, +0.002]	0/5	1.000
K7	color: HSV	0.985 ± 0.005	0.994	-0.012	[-0.019, -0.004]	0/5	0.176
K8	color: LAB	0.993 ± 0.009	0.994	-0.003	[-0.014, +0.008]	2/5	1.000
K9	color: CLAHE	0.994 ± 0.006	0.988	-0.002	[-0.008, +0.004]	0/5	1.000
R1	reference	0.943 ± 0.012	0.988	–	–	–	–
R2	size: stretch 224	0.948 ± 0.018	0.983	+0.005	[-0.010, +0.020]	3/5	1.000
R3	size: letterbox 384	0.960 ± 0.013	0.989	+0.017	[-0.001, +0.034]	5/5	0.829
R4	augmentation: none	0.934 ± 0.023	0.983	-0.009	[-0.029, +0.010]	2/5	1.000
R5	augmentation: strong	0.942 ± 0.014	0.989	-0.001	[-0.020, +0.017]	2/5	1.000
R6	color: grayscale	0.916 ± 0.019	0.971	-0.027	[-0.057, +0.003]	1/5	0.841
R7	color: HSV	0.908 ± 0.033	0.959	-0.035	[-0.067, -0.003]	0/5	0.569
R8	color: LAB	0.929 ± 0.017	0.953	-0.014	[-0.035, +0.006]	1/5	1.000

Pipeline	Change	CV macro-F1	Test	Δ	95% CI	Better	p_{Holm}
R9	color: CLAHE	0.950 ± 0.012	0.977	+0.007	[-0.005, +0.020]	4/5	1.000

Table S2: Background kept versus removed: CV macro-F1 of each paired pipeline, difference with 95% confidence interval, folds in which the kept version was better, and Holm-adjusted p (paired t-test).

Model	Pair	Kept	Removed	Δ [95% CI]	Folds	p_{Holm}
E1 All						
ResNet-50	K1 vs R1	0.995	0.954	+0.041 [+0.034, +0.047]	5/5	< 0.001
ResNet-50	K2 vs R2	0.996	0.960	+0.037 [+0.031, +0.043]	5/5	< 0.001
ResNet-50	K3 vs R3	0.998	0.962	+0.036 [+0.029, +0.043]	5/5	0.002
ResNet-50	K4 vs R4	0.986	0.946	+0.040 [+0.025, +0.056]	5/5	0.006
ResNet-50	K5 vs R5	0.994	0.952	+0.042 [+0.026, +0.058]	5/5	0.006
ResNet-50	K6 vs R6	0.986	0.929	+0.057 [+0.045, +0.069]	5/5	0.002
ResNet-50	K7 vs R7	0.987	0.920	+0.067 [+0.048, +0.086]	5/5	0.004
ResNet-50	K8 vs R8	0.991	0.940	+0.051 [+0.031, +0.071]	5/5	0.006
ResNet-50	K9 vs R9	0.993	0.955	+0.038 [+0.029, +0.046]	5/5	0.002
ConvNeXt-Tiny	K1 vs R1	0.994	0.961	+0.033 [+0.023, +0.043]	5/5	0.004
EfficientNet-B0	K1 vs R1	0.982	0.923	+0.059 [+0.038, +0.079]	5/5	0.005
ViT-S/16	K1 vs R1	0.995	0.957	+0.038 [+0.030, +0.046]	5/5	0.002
E2 Single						
ResNet-50	K1 vs R1	0.996	0.951	+0.045 [+0.035, +0.056]	5/5	0.002
ResNet-50	K2 vs R2	0.995	0.956	+0.040 [+0.032, +0.047]	5/5	0.001
ResNet-50	K3 vs R3	0.996	0.964	+0.032 [+0.014, +0.050]	5/5	0.012
ResNet-50	K4 vs R4	0.990	0.943	+0.047 [+0.022, +0.072]	5/5	0.012
ResNet-50	K5 vs R5	0.996	0.958	+0.038 [+0.029, +0.047]	5/5	0.002
ResNet-50	K6 vs R6	0.984	0.924	+0.060 [+0.050, +0.070]	5/5	0.001
ResNet-50	K7 vs R7	0.988	0.912	+0.075 [+0.057, +0.093]	5/5	0.002
ResNet-50	K8 vs R8	0.993	0.937	+0.056 [+0.038, +0.073]	5/5	0.004
ResNet-50	K9 vs R9	0.992	0.943	+0.049 [+0.038, +0.060]	5/5	0.002
ConvNeXt-Tiny	K1 vs R1	0.999	0.966	+0.033 [+0.025, +0.041]	5/5	0.002
EfficientNet-B0	K1 vs R1	0.974	0.904	+0.069 [+0.043, +0.095]	5/5	0.007
ViT-S/16	K1 vs R1	0.997	0.954	+0.044 [+0.024, +0.063]	5/5	0.011
E3 Multi						
ResNet-50	K1 vs R1	0.996	0.943	+0.053 [+0.041, +0.066]	5/5	0.003
ResNet-50	K2 vs R2	0.995	0.948	+0.047 [+0.030, +0.064]	5/5	0.009
ResNet-50	K3 vs R3	0.993	0.960	+0.033 [+0.017, +0.050]	5/5	0.019
ResNet-50	K4 vs R4	0.991	0.934	+0.057 [+0.031, +0.084]	5/5	0.019
ResNet-50	K5 vs R5	0.994	0.942	+0.052 [+0.035, +0.070]	5/5	0.009
ResNet-50	K6 vs R6	0.992	0.916	+0.076 [+0.054, +0.098]	5/5	0.006
ResNet-50	K7 vs R7	0.985	0.908	+0.077 [+0.039, +0.115]	5/5	0.019
ResNet-50	K8 vs R8	0.993	0.929	+0.064 [+0.047, +0.081]	5/5	0.005
ResNet-50	K9 vs R9	0.994	0.950	+0.044 [+0.032, +0.056]	5/5	0.005
ConvNeXt-Tiny	K1 vs R1	0.999	0.969	+0.030 [+0.015, +0.045]	5/5	0.019
EfficientNet-B0	K1 vs R1	0.964	0.888	+0.076 [+0.051, +0.101]	5/5	0.009
ViT-S/16	K1 vs R1	1.000	0.964	+0.036 [+0.019, +0.053]	5/5	0.019

Table S3: Per-class test F1 of the models trained with recipe F in each scenario.

Model	Churi	Dao	Khor	Kodal	Kural	Shabol
E1 All						
ConvNeXt-Tiny	1.000	0.994	1.000	0.992	1.000	1.000
ViT-S/16	1.000	1.000	1.000	1.000	1.000	1.000
ResNet-50	1.000	0.994	1.000	0.992	1.000	1.000
EfficientNet-B0	1.000	1.000	1.000	1.000	1.000	1.000
Test images (n)	68	79	62	67	63	80
E2 Single						
ConvNeXt-Tiny	0.986	1.000	1.000	0.988	1.000	1.000
ResNet-50	0.986	0.989	1.000	0.962	0.986	1.000
ViT-S/16	0.986	1.000	1.000	0.988	1.000	1.000

Model	Churi	Dao	Khor	Kodal	Kural	Shabol
EfficientNet-B0	1.000	1.000	1.000	1.000	1.000	1.000
Test images (n)	37	43	50	40	36	49
E3 Multi						
ViT-S/16	1.000	1.000	1.000	1.000	1.000	1.000
ConvNeXt-Tiny	1.000	1.000	1.000	1.000	1.000	1.000
ResNet-50	1.000	1.000	1.000	1.000	0.981	0.984
EfficientNet-B0	1.000	1.000	1.000	1.000	1.000	1.000
Test images (n)	31	36	12	27	27	31

Table S4: Transfer between single and multi images. Top: test macro-F1 on the same test images for the model trained on that image type (in-condition) and on the other type (cross-condition), with discordant pairs and Holm-adjusted exact McNemar p.

Model	Train → test	n	In-condition	Cross-condition	Discordant	p_{Holm}
ConvNeXt-Tiny	multi → single	255	0.996	0.858	36 / 1	< 0.001
ConvNeXt-Tiny	single → multi	164	1.000	0.831	30 / 0	< 0.001
EfficientNet-B0	multi → single	255	1.000	0.582	105 / 0	< 0.001
EfficientNet-B0	single → multi	164	1.000	0.642	61 / 0	< 0.001
ResNet-50	multi → single	255	0.987	0.542	109 / 1	< 0.001
ResNet-50	single → multi	164	0.994	0.768	43 / 1	< 0.001
ViT-S/16	multi → single	255	0.996	0.679	83 / 1	< 0.001
ViT-S/16	single → multi	164	1.000	0.669	55 / 0	< 0.001

Table S5: Per-class precision, recall and F1 under transfer between single and multi images.

Model	Train → test	Class	True	Predicted	Precision	Recall	F1
ConvNeXt-Tiny	single → multi	Churi	31	38	0.789	0.968	0.870
ConvNeXt-Tiny	single → multi	Dao	36	28	1.000	0.778	0.875
ConvNeXt-Tiny	single → multi	Khor	12	12	1.000	1.000	1.000
ConvNeXt-Tiny	single → multi	Kodal	27	31	0.806	0.926	0.862
ConvNeXt-Tiny	single → multi	Kural	27	36	0.611	0.815	0.698
ConvNeXt-Tiny	single → multi	Shabol	31	19	0.895	0.548	0.680
ConvNeXt-Tiny	multi → single	Churi	37	58	0.638	1.000	0.779
ConvNeXt-Tiny	multi → single	Dao	43	44	0.977	1.000	0.989
ConvNeXt-Tiny	multi → single	Khor	50	42	1.000	0.840	0.913
ConvNeXt-Tiny	multi → single	Kodal	40	47	0.787	0.925	0.851
ConvNeXt-Tiny	multi → single	Kural	36	32	0.906	0.806	0.853
ConvNeXt-Tiny	multi → single	Shabol	49	32	0.969	0.633	0.765
EfficientNet-B0	single → multi	Churi	31	38	0.789	0.968	0.870
EfficientNet-B0	single → multi	Dao	36	29	0.793	0.639	0.708
EfficientNet-B0	single → multi	Khor	12	12	1.000	1.000	1.000
EfficientNet-B0	single → multi	Kodal	27	41	0.537	0.815	0.647
EfficientNet-B0	single → multi	Kural	27	25	0.280	0.259	0.269
EfficientNet-B0	single → multi	Shabol	31	19	0.474	0.290	0.360
EfficientNet-B0	multi → single	Churi	37	102	0.363	1.000	0.532
EfficientNet-B0	multi → single	Dao	43	43	0.907	0.907	0.907
EfficientNet-B0	multi → single	Khor	50	20	1.000	0.400	0.571
EfficientNet-B0	multi → single	Kodal	40	39	0.692	0.675	0.684
EfficientNet-B0	multi → single	Kural	36	31	0.484	0.417	0.448
EfficientNet-B0	multi → single	Shabol	49	20	0.600	0.245	0.348
ResNet-50	single → multi	Churi	31	29	1.000	0.935	0.967
ResNet-50	single → multi	Dao	36	29	1.000	0.806	0.892
ResNet-50	single → multi	Khor	12	12	1.000	1.000	1.000
ResNet-50	single → multi	Kodal	27	25	0.720	0.667	0.692
ResNet-50	single → multi	Kural	27	48	0.375	0.667	0.480
ResNet-50	single → multi	Shabol	31	21	0.714	0.484	0.577
ResNet-50	multi → single	Churi	37	89	0.416	1.000	0.587
ResNet-50	multi → single	Dao	43	42	0.952	0.930	0.941

Model	Train → test	Class	True	Predicted	Precision	Recall	F1
ResNet-50	multi → single	Khor	50	20	1.000	0.400	0.571
ResNet-50	multi → single	Kodal	40	54	0.500	0.675	0.574
ResNet-50	multi → single	Kural	36	25	0.240	0.167	0.197
ResNet-50	multi → single	Shabol	49	25	0.560	0.286	0.378
ViT-S/16	single → multi	Churi	31	32	0.906	0.935	0.921
ViT-S/16	single → multi	Dao	36	42	0.762	0.889	0.821
ViT-S/16	single → multi	Khor	12	12	1.000	1.000	1.000
ViT-S/16	single → multi	Kodal	27	34	0.618	0.778	0.689
ViT-S/16	single → multi	Kural	27	32	0.281	0.333	0.305
ViT-S/16	single → multi	Shabol	31	12	0.500	0.194	0.279
ViT-S/16	multi → single	Churi	37	95	0.389	1.000	0.561
ViT-S/16	multi → single	Dao	43	42	0.976	0.953	0.965
ViT-S/16	multi → single	Khor	50	22	1.000	0.440	0.611
ViT-S/16	multi → single	Kodal	40	44	0.705	0.775	0.738
ViT-S/16	multi → single	Kural	36	31	0.677	0.583	0.627
ViT-S/16	multi → single	Shabol	49	21	0.952	0.408	0.571

Table S6: Capture-source check within the collection: test accuracy of the best E1 model per class and capture source, with Wilson 95% interval and Holm-adjusted Fisher exact p for the pooled minority sources against the main source.

Class	Source	Role	Images	Accuracy	Wilson 95% CI	p_{Holm}
Churi	Galaxy M14	main	315	1.000	[0.988, 1.000]	–
Churi	WhatsApp (no camera EXIF)	minority	2	1.000	[0.342, 1.000]	–
Churi	iPhone 14	minority	204	1.000	[0.982, 1.000]	–
Churi	all minority phones	minority (pooled)	206	1.000	[0.982, 1.000]	1.000
Dao	Redmi Note 15	minority	221	1.000	[0.983, 1.000]	–
Dao	WhatsApp (no camera EXIF)	minority	44	1.000	[0.920, 1.000]	–
Dao	iPhone 14	main	265	1.000	[0.986, 1.000]	–
Dao	all minority phones	minority (pooled)	265	1.000	[0.986, 1.000]	1.000
Khor	Galaxy M14	minority	223	1.000	[0.983, 1.000]	–
Khor	iPhone 14	main	248	1.000	[0.985, 1.000]	–
Khor	all minority phones	minority (pooled)	223	1.000	[0.983, 1.000]	1.000
Kodal	Redmi Note 15	minority	114	0.991	[0.952, 0.998]	–
Kodal	WhatsApp (no camera EXIF)	main	286	1.000	[0.987, 1.000]	–
Kodal	iPhone 14	minority	77	1.000	[0.953, 1.000]	–
Kodal	all minority phones	minority (pooled)	191	0.995	[0.971, 0.999]	1.000
Kural	Redmi Note 15	minority	158	1.000	[0.976, 1.000]	–
Kural	WhatsApp (no camera EXIF)	main	165	0.994	[0.967, 0.999]	–
Kural	iPhone 14	minority	154	1.000	[0.976, 1.000]	–
Kural	all minority phones	minority (pooled)	312	1.000	[0.988, 1.000]	1.000
Shabol	Redmi Note 15	minority	154	1.000	[0.976, 1.000]	–
Shabol	WhatsApp (no camera EXIF)	minority	149	1.000	[0.975, 1.000]	–
Shabol	iPhone 14	main	171	0.994	[0.968, 0.999]	–
Shabol	all minority phones	minority (pooled)	303	1.000	[0.988, 1.000]	1.000

Table S7: Strict capture-source test: models trained on iPhone 14 images only and tested on the iPhone test images and on all images from the other sources (macro-F1 and accuracy).

Run	Test images	n	Macro-F1	Accuracy
Recipe F	iPhone 14 test (same phone)	163	1.000	1.000
Recipe F	all other phones (unseen)	1831	0.534	0.556
Recipe F	Galaxy M14 (classes: Churi, Khor)	538	0.892	0.900
Recipe F	Redmi Note 15 (classes: Dao, Kodal, Kural, Shabol)	647	0.484	0.552
Recipe F	WhatsApp (no camera EXIF) (classes: Churi, Dao, Kodal, Kural, Shabol)	646	0.264	0.274
F, background removed	iPhone 14 test (same phone)	163	0.987	0.988

Run	Test images	n	Macro-F1	Accuracy
F, background removed	all other phones (unseen)	1831	0.595	0.606
F, background removed	Galaxy M14 (classes: Churi, Khor)	538	0.963	0.954
F, background removed	Redmi Note 15 (classes: Dao, Kodal, Kural, Shabol)	647	0.508	0.595
F, background removed	WhatsApp (no camera EXIF) (classes: Churi, Dao, Kodal, Kural, Shabol)	646	0.306	0.328
F + camera augmentation	iPhone 14 test (same phone)	163	1.000	1.000
F + camera augmentation	all other phones (unseen)	1831	0.507	0.532
F + camera augmentation	Galaxy M14 (classes: Churi, Khor)	538	0.885	0.894
F + camera augmentation	Redmi Note 15 (classes: Dao, Kodal, Kural, Shabol)	647	0.450	0.516
F + camera augmentation	WhatsApp (no camera EXIF) (classes: Churi, Dao, Kodal, Kural, Shabol)	646	0.242	0.246
DINOv2 + LR	iPhone 14 test (same phone)	163	0.990	0.994
DINOv2 + LR	all other phones (unseen)	1831	0.613	0.617
DINOv2 + LR	Galaxy M14 (classes: Churi, Khor)	538	0.963	0.957
DINOv2 + LR	Redmi Note 15 (classes: Dao, Kodal, Kural, Shabol)	647	0.486	0.594
DINOv2 + LR	WhatsApp (no camera EXIF) (classes: Churi, Dao, Kodal, Kural, Shabol)	646	0.332	0.358

Table S8: Calibration of the E1 models on the test set before and after temperature scaling (T fitted on cross-validation predictions).

Model	T	Conf.	Acc.	ECE	ECE_T	NLL	NLL_T	Brier	Brier_T
ConvNeXt-Tiny	0.39	0.913	0.998	0.086	0.002	0.095	0.010	0.012	0.004
ViT-S/16	0.36	0.912	1.000	0.088	0.001	0.093	0.002	0.011	0.001
ResNet-50	0.35	0.909	0.998	0.090	0.004	0.102	0.015	0.016	0.006
EfficientNet-B0	0.37	0.887	1.000	0.113	0.003	0.125	0.004	0.024	0.001

Table S9: Per-class share of Grad-CAM attention on the weapon region for the pipelines K1, F and R1 (fold-0 models).

Class	K1	F	R1
E1 All (ConvNeXt-Tiny)			
Churi	0.295	0.359	0.346
Dao	0.377	0.470	0.368
Khor	0.269	0.293	0.261
Kodal	0.345	0.410	0.372
Kural	0.251	0.326	0.289
Shabol	0.184	0.214	0.260
E2 Single (ConvNeXt-Tiny)			
Churi	0.258	0.304	0.304
Dao	0.309	0.360	0.286
Khor	0.259	0.326	0.262
Kodal	0.311	0.349	0.336
Kural	0.246	0.250	0.282
Shabol	0.134	0.174	0.240
E3 Multi (ViT-S/16)			
Churi	0.079	0.068	0.395
Dao	0.195	0.087	0.312
Khor	0.108	0.031	0.290
Kodal	0.068	0.051	0.391
Kural	0.035	0.025	0.192
Shabol	0.101	0.051	0.181

Table S10: Preprocessing check on three architectures (E1): CV macro-F1 of the settings that determined recipe F, difference from each model's reference K1, folds in which the setting was better, Holm-adjusted p over the six new comparisons (paired t-test; ResNet-50 rows are reproduced from Table S1 for reference), and test macro-F1.

Model	Pipeline	Setting	CV macro-F1	Δ	Better	p_{Holm}	Test
ResNet-50	K1	Reference (letterbox 224, RGB)	0.995 ± 0.003	–	–	–	0.995
ResNet-50	K3	Letterbox 384	0.998 ± 0.002	+0.003	4/5	–	0.998
ResNet-50	K6	Grayscale	0.986 ± 0.004	-0.009	0/5	–	0.995
ResNet-50	K7	HSV	0.987 ± 0.002	-0.008	0/5	–	0.992
ConvNeXt-Tiny	K1	Reference (letterbox 224, RGB)	0.994 ± 0.003	–	–	–	0.998
ConvNeXt-Tiny	K3	Letterbox 384	0.998 ± 0.002	+0.004	4/5	0.094	0.998
ConvNeXt-Tiny	K6	Grayscale	0.988 ± 0.002	-0.007	0/5	0.041	0.995
ConvNeXt-Tiny	K7	HSV	0.992 ± 0.003	-0.002	2/5	0.658	0.993
ViT-S/16	K1	Reference (letterbox 224, RGB)	0.995 ± 0.002	–	–	–	0.995
ViT-S/16	K3	Letterbox 384	0.996 ± 0.005	+0.001	3/5	0.706	0.998
ViT-S/16	K6	Grayscale	0.983 ± 0.007	-0.012	0/5	0.094	0.993
ViT-S/16	K7	HSV	0.991 ± 0.009	-0.004	2/5	0.658	0.990

Table S11: Images without weapons by model and image type: number of images, false alarms at the threshold that retains 95% of the BD-LW test weapons, mean calibrated confidence and share with confidence of at least 0.90; the right-hand columns give the weapon class assigned to all 112 images.

Model	Type	n	FA	Mean conf.	≥ 0.90	Churi	Dao	Khor	Kodal	Kural	Shabol
ResNet-50	Look-alike	72	1	0.588	0.069	54	13	7	17	1	20
ResNet-50	Everyday	40	0	0.682	0.200						
EfficientNet-B0	Look-alike	72	0	0.545	0.014	41	62	3	2	2	2
EfficientNet-B0	Everyday	40	0	0.591	0.075						
ConvNeXt-Tiny	Look-alike	72	0	0.639	0.153	40	12	7	23	6	24
ConvNeXt-Tiny	Everyday	40	0	0.675	0.300						
ViT-S/16	Look-alike	72	1	0.611	0.097	36	33	4	10	6	23
ViT-S/16	Everyday	40	0	0.769	0.450						

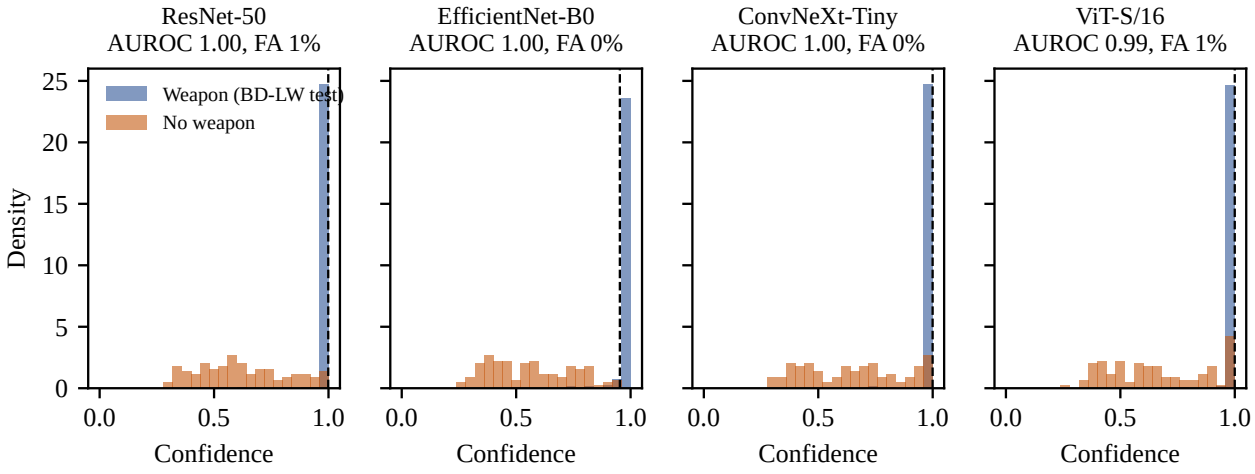


Fig. S9 Calibrated confidence of the 419 BD-LW test weapons (blue) and the 112 images without weapons (orange) for each E1 model; the dashed line marks the threshold that retains 95% of the weapons (FA: false-alarm rate at this threshold).

Table S12: Preprocessing settings. Each setting changes one factor (bold) from the reference K1; every setting is run with the background kept (K) and removed (R).

Setting	Geometry	Augmentation	Colour
K1 / R1	letterbox 224	light	RGB
K2 / R2	**stretch 224**	light	RGB
K3 / R3	**letterbox 384**	light	RGB
K4 / R4	letterbox 224	**none**	RGB
K5 / R5	letterbox 224	**strong**	RGB
K6 / R6	letterbox 224	light	**grayscale**
K7 / R7	letterbox 224	light	**HSV**
K8 / R8	letterbox 224	light	**CIELAB**
K9 / R9	letterbox 224	light	**CLAHE RGB**

Table S13: CV macro-F1 of ResNet-50 for each preprocessing setting with the background kept (K) and removed (R). Asterisks mark a significant difference from the reference K1 or R1 (paired t-test, Holm-adjusted $p < 0.05$); bold marks the best K value per scenario.

	E1 All		E2 Single		E3 Multi	
Setting	K	R	K	R	K	R
1 Reference	0.995	0.954	**0.996**	0.951	**0.996**	0.943
2 Stretch 224	0.996	0.960	0.995	0.956	0.995	0.948
3 Letterbox 384	**0.998**	0.962	**0.996**	0.964	0.993	0.960
4 No augmentation	0.986*	0.946	0.990	0.943	0.991	0.934
5 Strong augmentation	0.994	0.952	**0.996**	0.958	0.994	0.942
6 Grayscale	0.986	0.929*	0.984*	0.924	0.992	0.916
7 HSV	0.987	0.920	0.988*	0.913*	0.985	0.908
8 CIELAB	0.991	0.940	0.993	0.937	0.993	0.929
9 CLAHE RGB	0.993	0.955	0.992	0.943	0.994	0.950

Table S14: Model size, computation and inference time at the input size of recipe F. Training time is per image and epoch in E1, derived from Table **??**.

Model	Params (M)	GFLOPs /image	CPU (ms/image)	GPU (ms/image)	GPU throughput (images/s)	Training (ms/image/epoch)
ResNet-50	23.5	24.0	164	6.4	216	14.6
EfficientNet-B0	4.0	2.3	56	8.2	543	9.5
ConvNeXt-Tiny	27.8	26.2	153	7.2	210	15.3
ViT-S/16	21.8	24.8	181	6.1	298	12.3
DINOv2 + LR	21.6	11.0	77	6.2	746	–

Table S15: Size distribution of the 2,501 similarity groups: number of groups of each size, number of groups and images, size of the largest group, and share of images that are in a group with at least one other image. Three groups link images of two classes, so the per-class group counts add up to 2,504.

Class	1	2	3	4–5	6–10	>10	Groups	Images	Largest	Share in groups
All	2281	137	35	33	9	6	2501	2950	21	0.227
Churi	393	41	8	5	0	0	447	521	5	0.246
Dao	448	30	2	4	0	0	484	530	4	0.155
Khor	231	25	11	12	4	5	288	471	21	0.510
Kodal	452	9	1	1	0	0	463	477	4	0.052
Kural	383	18	3	6	2	1	413	477	12	0.197
Shabol	380	11	10	5	3	0	409	474	9	0.198

Table S16: Image corruptions used in the robustness evaluation, applied to the letterboxed 384×384 model input at severity levels 1 to 5. One random generator per corruption and level (seed 42+level) sets the noise and the position of the occluding square.

Corruption	1	2	3	4	5
Gaussian blur, σ (px at 224 px, scaled by 384/224)	0.5	1	2	3	4
Darkness, brightness factor	0.8	0.6	0.45	0.3	0.2
JPEG compression, quality	60	40	25	15	8
Gaussian noise, SD (of 255)	8	16	24	32	48
Occlusion, grey square (share of image area)	0.05	0.10	0.20	0.30	0.40
Rotation (degrees, grey fill)	10	20	30	45	60

Table S17: Checks used to validate the BD-LW annotations: number of images flagged by each check and number of labels changed after review.

Check	What is verified	Flagged	Changed
Double annotation, class	Two annotators assign the same class (300 images; $\kappa = 0.976$)	6	0
Double annotation, subset	Two annotators agree on *single* or *multi* (300 images; $\kappa = 0.980$)	3	0
Neighbour consistency	Nearest neighbours in feature space share the label	26	0
Class centroid distance	Image is not an outlier of its class	37	0
Post-hoc error review	Misclassified test images are not label errors	4	0

Table S18: Training settings shared by all runs.

Setting	Value
Optimizer	AdamW [**?**], weight decay 0.05
Learning rate	2×10^{-4}, 1-epoch warm-up, cosine decay [**?**]
Batch size, epochs	32, 25
Loss	cross-entropy, label smoothing 0.1 [**?**]
Precision	mixed (FP16)
Cross-validation	5 folds, seed 42

Table S19: Grad-CAM share on the weapon of the E1 models per class and per subset (mean over the five folds and all test images with a mask).

Model	Churi	Dao	Khor	Kodal	Kural	Shabol	Single	Multi
ResNet-50	0.330	0.387	0.298	0.335	0.249	0.182	0.267	0.348
EfficientNet-B0	0.245	0.301	0.194	0.244	0.177	0.177	0.204	0.261
ConvNeXt-Tiny	0.334	0.469	0.308	0.393	0.316	0.234	0.307	0.406
ViT-S/16	0.102	0.104	0.091	0.161	0.079	0.090	0.090	0.129

Table S20: Grad-CAM share on the weapon of ConvNeXt-Tiny and ViT-S/16 in all three scenarios for pipelines K1, F and R1 (fold-0 models), with ρ in parentheses (ρ = median ratio of share to weapon area; $\rho > 1$ means attention is concentrated on the weapon).

Scenario (n)	Model	Weapon area	K1 (kept)	F (final)	R1 (removed)
E1 All (406)	ConvNeXt-Tiny	0.088	0.288 (3.8)	0.347 (4.8)	0.318 (4.5)
	ViT-S/16		0.078 (0.8)	0.105 (0.8)	0.268 (3.2)
E2 Single (250)	ConvNeXt-Tiny	0.074	0.250 (4.4)	0.293 (5.2)	0.283 (5.0)
	ViT-S/16		0.103 (1.1)	0.079 (0.9)	0.244 (3.5)
E3 Multi (156)	ConvNeXt-Tiny	0.112	0.346 (3.2)	0.424 (4.1)	0.390 (3.9)
	ViT-S/16		0.103 (0.8)	0.057 (0.4)	0.299 (2.9)

Table S21: Zero-shot CLIP on the 419 BD-LW test images: macro-F1, accuracy and recall per class for each model and prompt set. Prompts: English name (knife, machete, straight razor, hoe, axe, iron crowbar); the same name in six templates; or the local name with a short description.

Model	Prompts	Macro-F1	Acc.	Churi	Dao	Khor	Kodal	Kural	Shabol
ViT-B-32 (laion2b_s34b_b79k)	Name	0.448	0.415	0.53	0.21	0.69	0.55	0.32	0.26
ViT-B-32 (laion2b_s34b_b79k)	Name, 6 templates	0.437	0.401	0.53	0.20	0.69	0.55	0.32	0.20
ViT-B-32 (laion2b_s34b_b79k)	Local name + description	0.549	0.554	0.77	0.27	0.74	0.45	0.30	0.80
ViT-B-16 (openai)	Name	0.316	0.332	0.78	0.32	0.21	0.04	0.40	0.25
ViT-B-16 (openai)	Name, 6 templates	0.317	0.329	0.77	0.33	0.26	0.04	0.35	0.24
ViT-B-16 (openai)	Local name + description	0.506	0.496	0.81	0.37	0.82	0.09	0.57	0.39
ViT-L-14 (openai)	Name	0.586	0.582	0.54	0.46	0.85	0.19	0.70	0.76
ViT-L-14 (openai)	Name, 6 templates	0.633	0.616	0.56	0.49	0.94	0.40	0.68	0.66
ViT-L-14 (openai)	Local name + description	0.707	0.704	0.82	0.39	1.00	0.58	0.79	0.71
ViT-L-14 (laion2b_s32b_b82k)	Name	0.550	0.546	0.52	0.15	0.89	0.88	0.25	0.65
ViT-L-14 (laion2b_s32b_b82k)	Name, 6 templates	0.560	0.554	0.56	0.15	0.94	0.90	0.32	0.55
ViT-L-14 (laion2b_s32b_b82k)	Local name + description	0.666	0.661	0.82	0.38	0.94	0.81	0.38	0.69

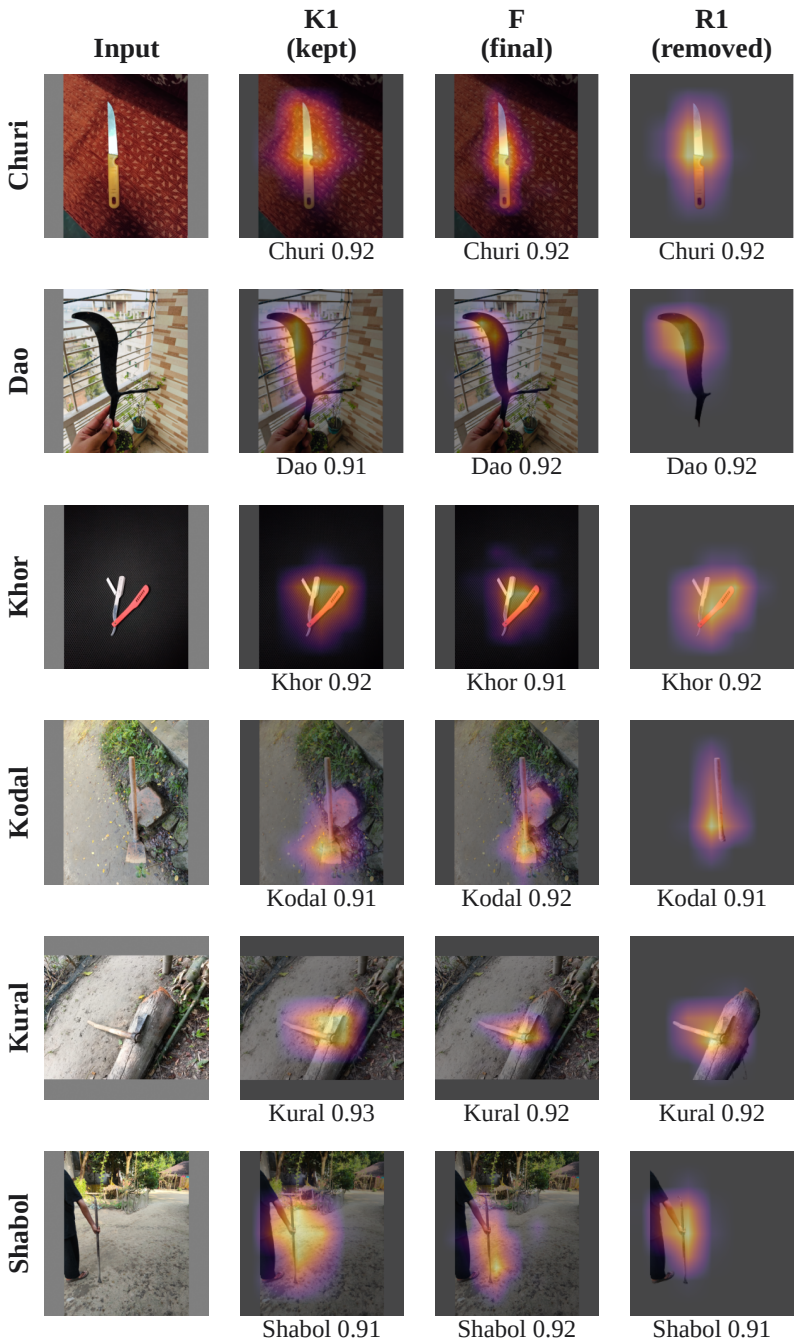


Fig. S11 Grad-CAM maps of ConvNeXt-Tiny (E1, first-fold models) for one test image per class and pipelines K1, F and R1, with predicted class and confidence. Brighter regions contribute more to the prediction.

Table S22: Cross-validation comparisons with a correction for dependent folds. Δ: mean difference in macro-F1 over the five folds (first minus second setting); Folds: folds in which the first setting was better; p_{Holm} of the paired t-test (indicative); p and Holm-adjusted p of the corrected resampled t-test (Nadeau and Bengio, 2003), Holm within each family and scenario.

Family	Comparison	Δ	Folds	p_{Holm}, t	p, corr.	p_{Holm}, corr.
E1						
ablation	ResNet-50 K2 vs K1	+0.002	4/5	0.864	0.224	1.000
ablation	ResNet-50 K3 vs K1	+0.003	4/5	0.864	0.234	1.000
ablation	ResNet-50 K4 vs K1	-0.009	0/5	0.015	0.004	0.069
ablation	ResNet-50 K5 vs K1	-0.001	1/5	1.000	0.769	1.000
ablation	ResNet-50 K6 vs K1	-0.009	0/5	0.154	0.050	0.546
ablation	ResNet-50 K7 vs K1	-0.008	0/5	0.073	0.021	0.294
ablation	ResNet-50 K8 vs K1	-0.004	1/5	0.654	0.168	1.000
ablation	ResNet-50 K9 vs K1	-0.002	1/5	0.864	0.363	1.000
ablation	ResNet-50 R2 vs R1	+0.005	4/5	0.864	0.289	1.000
ablation	ResNet-50 R3 vs R1	+0.008	5/5	0.133	0.041	0.488
ablation	ResNet-50 R4 vs R1	-0.009	1/5	0.864	0.322	1.000
ablation	ResNet-50 R5 vs R1	-0.003	3/5	1.000	0.753	1.000
ablation	ResNet-50 R6 vs R1	-0.026	0/5	0.015	0.004	0.069
ablation	ResNet-50 R7 vs R1	-0.035	0/5	0.091	0.027	0.355
ablation	ResNet-50 R8 vs R1	-0.014	1/5	0.864	0.222	1.000
ablation	ResNet-50 R9 vs R1	+0.001	3/5	1.000	0.809	1.000
background	ViT-S/16 K1 vs R1	+0.038	5/5	0.002	< 0.001	0.008
background	ResNet-50 K1 vs R1	+0.041	5/5	< 0.001	< 0.001	0.004
background	ResNet-50 K2 vs R2	+0.037	5/5	< 0.001	< 0.001	0.005
background	ResNet-50 K3 vs R3	+0.036	5/5	0.002	< 0.001	0.008
background	ResNet-50 K4 vs R4	+0.040	5/5	0.006	0.009	0.025
background	ResNet-50 K5 vs R5	+0.042	5/5	0.006	0.008	0.025
background	ResNet-50 K6 vs R6	+0.057	5/5	0.002	< 0.001	0.008
background	ResNet-50 K7 vs R7	+0.067	5/5	0.004	0.003	0.017
background	ResNet-50 K8 vs R8	+0.051	5/5	0.006	0.009	0.025
background	ResNet-50 K9 vs R9	+0.038	5/5	0.002	0.001	0.009
background	EfficientNet-B0 K1 vs R1	+0.059	5/5	0.005	0.006	0.024
background	ConvNeXt-Tiny K1 vs R1	+0.033	5/5	0.004	0.004	0.018
models	ViT-S/16 vs ConvNeXt-Tiny (F)	-0.000	2/5	0.742	0.825	0.825
models	ResNet-50 vs ConvNeXt-Tiny (F)	-0.001	0/5	0.132	0.170	0.339
models	EfficientNet-B0 vs ConvNeXt-Tiny (F)	-0.006	0/5	0.046	0.054	0.161
E2						
ablation	ResNet-50 K2 vs K1	-0.001	0/5	1.000	0.533	1.000
ablation	ResNet-50 K3 vs K1	-0.000	1/5	1.000	0.984	1.000
ablation	ResNet-50 K4 vs K1	-0.006	0/5	0.450	0.150	1.000
ablation	ResNet-50 K5 vs K1	+0.000	1/5	1.000	0.978	1.000
ablation	ResNet-50 K6 vs K1	-0.012	0/5	0.020	0.006	0.089
ablation	ResNet-50 K7 vs K1	-0.008	0/5	0.019	0.006	0.085
ablation	ResNet-50 K8 vs K1	-0.003	0/5	0.302	0.086	0.952
ablation	ResNet-50 K9 vs K1	-0.004	0/5	0.302	0.088	0.952
ablation	ResNet-50 R2 vs R1	+0.005	5/5	0.312	0.104	0.952
ablation	ResNet-50 R3 vs R1	+0.013	4/5	0.544	0.205	1.000
ablation	ResNet-50 R4 vs R1	-0.008	1/5	0.818	0.370	1.000
ablation	ResNet-50 R5 vs R1	+0.007	4/5	0.544	0.191	1.000
ablation	ResNet-50 R6 vs R1	-0.027	0/5	0.075	0.023	0.298
ablation	ResNet-50 R7 vs R1	-0.038	0/5	0.015	0.004	0.071
ablation	ResNet-50 R8 vs R1	-0.013	0/5	0.116	0.036	0.436
ablation	ResNet-50 R9 vs R1	-0.007	1/5	0.544	0.191	1.000
background	ViT-S/16 K1 vs R1	+0.044	5/5	0.011	0.015	0.044
background	ResNet-50 K1 vs R1	+0.045	5/5	0.002	0.001	0.011
background	ResNet-50 K2 vs R2	+0.040	5/5	0.001	< 0.001	0.006
background	ResNet-50 K3 vs R3	+0.032	5/5	0.012	0.029	0.049
background	ResNet-50 K4 vs R4	+0.047	5/5	0.012	0.024	0.049
background	ResNet-50 K5 vs R5	+0.038	5/5	0.002	0.001	0.011
background	ResNet-50 K6 vs R6	+0.060	5/5	0.001	< 0.001	0.005
background	ResNet-50 K7 vs R7	+0.075	5/5	0.002	0.001	0.011
background	ResNet-50 K8 vs R8	+0.056	5/5	0.004	0.004	0.020

Family	Comparison	Δ	Folds	p_{Holm}, t	p, corr.	p_{Holm}, corr.
background	ResNet-50 K9 vs R9	+0.049	5/5	0.002	0.001	0.011
background	EfficientNet-B0 K1 vs R1	+0.069	5/5	0.007	0.008	0.031
background	ConvNeXt-Tiny K1 vs R1	+0.033	5/5	0.002	0.002	0.011
models	ViT-S/16 vs ConvNeXt-Tiny (F)	-0.001	0/5	0.748	0.541	1.000
models	ResNet-50 vs ConvNeXt-Tiny (F)	-0.001	0/5	0.748	0.541	1.000
models	EfficientNet-B0 vs ConvNeXt-Tiny (F)	-0.006	0/5	0.410	0.283	0.850
E3						
ablation	ResNet-50 K2 vs K1	-0.001	1/5	1.000	0.715	1.000
ablation	ResNet-50 K3 vs K1	-0.003	1/5	1.000	0.443	1.000
ablation	ResNet-50 K4 vs K1	-0.005	1/5	1.000	0.364	1.000
ablation	ResNet-50 K5 vs K1	-0.002	1/5	1.000	0.361	1.000
ablation	ResNet-50 K6 vs K1	-0.005	0/5	1.000	0.243	1.000
ablation	ResNet-50 K7 vs K1	-0.012	0/5	0.176	0.041	0.648
ablation	ResNet-50 K8 vs K1	-0.003	2/5	1.000	0.607	1.000
ablation	ResNet-50 K9 vs K1	-0.002	0/5	1.000	0.541	1.000
ablation	ResNet-50 R2 vs R1	+0.005	3/5	1.000	0.584	1.000
ablation	ResNet-50 R3 vs R1	+0.017	5/5	0.829	0.157	1.000
ablation	ResNet-50 R4 vs R1	-0.009	2/5	1.000	0.432	1.000
ablation	ResNet-50 R5 vs R1	-0.001	2/5	1.000	0.901	1.000
ablation	ResNet-50 R6 vs R1	-0.027	1/5	0.841	0.167	1.000
ablation	ResNet-50 R7 vs R1	-0.035	0/5	0.569	0.112	1.000
ablation	ResNet-50 R8 vs R1	-0.014	1/5	1.000	0.260	1.000
ablation	ResNet-50 R9 vs R1	+0.007	4/5	1.000	0.356	1.000
background	ViT-S/16 K1 vs R1	+0.036	5/5	0.019	0.017	0.081
background	ResNet-50 K1 vs R1	+0.053	5/5	0.003	0.001	0.016
background	ResNet-50 K2 vs R2	+0.047	5/5	0.009	0.006	0.039
background	ResNet-50 K3 vs R3	+0.033	5/5	0.019	0.019	0.081
background	ResNet-50 K4 vs R4	+0.057	5/5	0.019	0.016	0.081
background	ResNet-50 K5 vs R5	+0.052	5/5	0.009	0.005	0.038
background	ResNet-50 K6 vs R6	+0.076	5/5	0.006	0.003	0.028
background	ResNet-50 K7 vs R7	+0.077	5/5	0.019	0.020	0.081
background	ResNet-50 K8 vs R8	+0.064	5/5	0.005	0.002	0.023
background	ResNet-50 K9 vs R9	+0.044	5/5	0.005	0.003	0.025
background	EfficientNet-B0 K1 vs R1	+0.076	5/5	0.009	0.005	0.038
background	ConvNeXt-Tiny K1 vs R1	+0.030	5/5	0.019	0.020	0.081
models	ResNet-50 vs ViT-S/16 (F)	-0.002	0/5	0.356	0.338	0.675
models	EfficientNet-B0 vs ViT-S/16 (F)	-0.008	0/5	0.028	0.035	0.106
models	ConvNeXt-Tiny vs ViT-S/16 (F)	-0.001	0/5	0.356	0.477	0.675

S4 Reproducibility: which notebook produces each result

All notebooks run on Kaggle (one Tesla T4 GPU); the inputs of each are listed in its first cell. Run them in the order below; each reads the outputs of the earlier ones.

Notebook	Results
0 Stage 0: data curation	standardization, near-duplicates, similarity groups, split; Fig. 3; Tables S15, S17 (checks 3–4)
1 Background removal (IS-Net)	masks and cut-outs; Fig. 2; Supplementary Fig. S1
2 Training (scenarios all, single, multi)	all 435 networks; Table 2 (training time); Table S18
3 Analysis	Tables 2, 4, S1–S9, S12–S14, S16; Figs. 4, 5
4 Paper figures	Figs. 1, 4–7; Supplementary Figs. S2–S8, S10
5 Grad-CAM, same architectures	Table S20; Fig. S11
6 Ablation check	Table S10
7 Images without weapons	Table S11; Supplementary Fig. S9
8 Annotation validation	Table S17 (checks 1, 2 and 5)
9 Robustness, five folds	Fig. 6
10 Grad-CAM, all models	Table 3; Fig. 7; Table S19
11 Zero-shot CLIP	Table S21
12 Merge and statistics	Fig. 6; Tables S15, S22
13 Misclassified test images	Supplementary Fig. S10b